

EXPERIMENT 7 – REPRODUCING AN IMAGE USING PROMPTS FOR IMAGE GENERATION

AIM:

TO DEMONSTRATE THE ABILITY OF TEXT-TO-IMAGE GENERATION TOOLS TO REPRODUCE AN EXISTING IMAGE BY CRAFTING PRECISE PROMPTS.

OBJECTIVE:

TO IDENTIFY KEY ELEMENTS WITHIN A REFERENCE IMAGE AND USE PROMPT ENGINEERING TECHNIQUES TO GENERATE AN IMAGE AS CLOSE AS POSSIBLE TO THE ORIGINAL IMAGE USING AI IMAGE GENERATION TOOLS.

AI TOOLS USED:

- CHATGPT IMAGE GENERATION
- GEMINI
- BING IMAGE CREATOR
- LEONARDO AI
- CANVA AI
- MIDJOURNEY

REFERENCE IMAGE 1: BUTTERFLY COLORING PAGE

STEP 1: IMAGE ANALYSIS

I) OBJECTS/SUBJECT

- LARGE BUTTERFLY AT THE CENTER
- SYMMETRICAL WINGS
- DECORATIVE WING PATTERNS
- SIMPLE OUTLINE DRAWING

II) BACKGROUND

- PLAIN WHITE BACKGROUND

- NO ADDITIONAL OBJECTS

III) LIGHTING

- FLAT LIGHTING
- NO SHADOWS

IV) STYLE

- BLACK-AND-WHITE COLORING BOOK ILLUSTRATION
- CLEAN VECTOR LINE ART
- CARTOON STYLE

V) COLOR COMBINATION

- MONOCHROME (BLACK OUTLINES ON WHITE BACKGROUND)

VI) CAMERA QUALITY

- HIGH-RESOLUTION DIGITAL ILLUSTRATION
- SHARP EDGES

INITIAL PROMPT:

"DRAW A BUTTERFLY COLORING PAGE WITH LARGE WINGS AND SIMPLE BLACK OUTLINES ON A WHITE BACKGROUND."

OBSERVATION

THE GENERATED IMAGE CONTAINED A BUTTERFLY BUT LACKED SYMMETRY AND DETAILED WING PATTERNS.

REFINED PROMPT:

"CREATE A HIGH-RESOLUTION BLACK-AND-WHITE COLORING BOOK PAGE FEATURING A LARGE SYMMETRICAL BUTTERFLY CENTERED ON A PLAIN WHITE BACKGROUND. USE CLEAN VECTOR-STYLE LINE ART WITH BOLD OUTLINES, DECORATIVE WING PATTERNS, AND NO SHADING OR COLORS. THE IMAGE SHOULD LOOK LIKE A PRINTABLE CHILDREN'S COLORING PAGE WITH SHARP DETAILS AND BALANCED WING DESIGN."

OBSERVATION

THE GENERATED IMAGE CLOSELY MATCHED THE REFERENCE IMAGE. WING SYMMETRY, OUTLINE QUALITY, AND COLORING-BOOK STYLE IMPROVED SIGNIFICANTLY.

REFERENCE IMAGE 2: PRODUCT LAUNCH POSTER

STEP 1: IMAGE ANALYSIS:

I) OBJECTS/SUBJECT

- **PRODUCT LAUNCH ANNOUNCEMENT POSTER**
- **MODERN TYPOGRAPHY**
- **PROMOTIONAL DESIGN ELEMENTS**
- **PRODUCT SHOWCASE SECTION**

II) BACKGROUND

- **CLEAN PROFESSIONAL LAYOUT**
- **GRADIENT AND GEOMETRIC SHAPES**

III) LIGHTING

- **BRIGHT COMMERCIAL DESIGN APPEARANCE**

IV) STYLE

- **MODERN MARKETING POSTER**
- **CORPORATE ADVERTISING STYLE**

V) COLOR COMBINATION

- **BLUE, PURPLE, PINK GRADIENTS**
- **WHITE TEXT**
- **HIGH CONTRAST DESIGN**

VI) CAMERA QUALITY

- **PROFESSIONAL GRAPHIC DESIGN QUALITY**
- **HIGH-RESOLUTION POSTER**

INITIAL PROMPT:

"CREATE A MODERN PRODUCT LAUNCH POSTER WITH A PRODUCT IMAGE AND PROMOTIONAL TEXT."

OBSERVATION

THE GENERATED OUTPUT RESEMBLED A POSTER BUT LACKED PROFESSIONAL LAYOUT AND VISUAL APPEAL.

REFINED PROMPT:

"DESIGN A PROFESSIONAL PRODUCT LAUNCH PROMOTIONAL POSTER IN PORTRAIT ORIENTATION. USE A MODERN CORPORATE MARKETING STYLE WITH VIBRANT BLUE, PURPLE, AND PINK GRADIENT BACKGROUNDS. INCLUDE BOLD HEADLINE TEXT READING 'PRODUCT LAUNCH', CLEAN TYPOGRAPHY, GEOMETRIC DESIGN ELEMENTS, PRODUCT SHOWCASE AREA, HIGH CONTRAST WHITE TEXT, SMOOTH LIGHTING EFFECTS, PREMIUM BRANDING APPEARANCE, AND HIGH-RESOLUTION COMMERCIAL ADVERTISING QUALITY."

OBSERVATION

THE GENERATED POSTER CLOSELY RESEMBLED THE ORIGINAL DESIGN. TYPOGRAPHY, COLOR SCHEME, AND OVERALL COMPOSITION WERE SIGNIFICANTLY IMPROVED.

COMPARISON OF RESULTS:

FEATURE	INITIAL OUTPUT	REFINED OUTPUT
OBJECT ACCURACY	MODERATE	HIGH
BACKGROUND SIMILARITY	LOW	HIGH
STYLE MATCHING	MODERATE	VERY HIGH
COLOR REPRODUCTION	LOW	HIGH
DETAIL LEVEL	BASIC	ADVANCED
OVERALL SIMILARITY	60%	90%+

OUTPUT:

THE REFINED PROMPTS GENERATED IMAGES THAT WERE MUCH CLOSER TO THE ORIGINAL REFERENCE IMAGES. DETAILED PROMPT ENGINEERING INVOLVING OBJECT DESCRIPTION, STYLE, BACKGROUND, LIGHTING, COLOR COMBINATIONS, AND COMPOSITION SIGNIFICANTLY IMPROVED IMAGE QUALITY AND SIMILARITY.

RESULT:

THUS, THE ORIGINAL IMAGES WERE SUCCESSFULLY REPRODUCED USING PROMPT ENGINEERING TECHNIQUES AND AI IMAGE GENERATION TOOLS. THE EXPERIMENT DEMONSTRATED THAT DETAILED AND WELL-STRUCTURED PROMPTS GREATLY ENHANCE THE ACCURACY AND QUALITY OF AI-GENERATED IMAGES.